

Spatial Modelling of Satellite Data for the Development of a Wildfire Intensity Index in Italy

Faculty of Information Engineering, Computer Science and Statistics
Master's Degree in Statistical Methods and Applications

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SAPIENZA
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Why study wildfires?

1 Introduction

- Major environmental and socio-economic challenge, affecting ecosystems, human activities and landscapes;
- Italy among the most exposed European countries;
- complex phenomenon resulting from the interaction of climatic, environmental and anthropogenic factors;
- context of climate change that increments favorable condition for fire ignitions and spread.

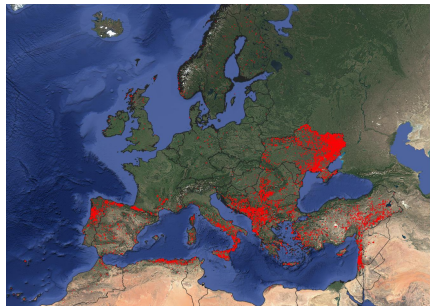


Figure: Burnt scars produced by wildfires during the 2024 fire season



Research objectives

1 Introduction

The main objective of this thesis is to develop a spatially explicit model of **wildfire intensity** in Italy, which allows to:

- estimate spatial heterogeneity in wildfire occurrence;
- relate wildfire intensity to environmental, climatic and anthropogenic drivers;
- account for residual spatial dependence;
- provide a basis for a wildfire intensity index.



Burnt Areas Database - Italian Terrestrial Ecosystems (BAD-ITE)

2 Data sources and Pre-processing

- BAD-ITE dataset derived from EFFIS satellite burned-area polygons;
- period considered: 2018–2025;
- aggregation by EFFIS_ID to obtain one record per wildfire event;
- polygon centroids used as point locations for modelling;
- study domain restricted to natural and semi-natural areas;
- exclusion of minor islands;
- final dataset: 9561 wildfire events.





Explanatory covariates

2 Data sources and Pre-processing

The objective of the model is also to investigate the relationship between wildfire occurrence and a set of environmental and anthropogenic predictors:

Category	Variables Included
Landscape	Elevation, latitude, distance from coast
Climate	Temperature, Land Surface Temperature, annual/monthly precipitation, solar radiation
Vegetation	Tree cover density
Human	Population, human pressure index

All raster covariates were harmonized and standardized before model fitting.



Exploratory Data Analysis: Wildfire Patterns

3 Exploratory Data Analysis

Key National Patterns:

- **Number of wildfires:** The number of wildfires shows an overall increasing pattern over the study period.
- **Total burned area:** Burned area is driven by extreme fire seasons (e.g., 2021 accounts for 28.5% of overall total burned area).

Ecological Heterogeneity:

- Strong imbalance across ecoregions, with the **Tyrrhenian ecoregion** recording over 75% of all events.
- **Agricultural and Forest** ecosystems account for the vast majority of ignitions.

Ecoregions in Italy





Why a spatial statistical model?

4 Methods

Wildfires are spatially structured events.

Nearby locations tend to share similar:

- climatic conditions
- vegetation characteristics
- topography
- anthropogenic factors

Therefore, wildfire occurrences are **not spatially independent**.

A suitable model should account for:

- observed environmental drivers
- residual spatial dependence



Why LGCP?

4 Methods

Point Patterns

Wildfires are modelled as point patterns.



Inhomogeneous Spatial Poisson Process

Flexible but assumes no residual spatial dependence.



Log-Gaussian Cox Processes (LGCP)

Special case of Cox process where log-intensity is driven by a **Gaussian Random Field**

$$\log \lambda(s) = \beta_0 + \sum_{k=1}^p \beta_k x_k(s) + \sum_{j=1}^q f_j(z_j(s)) + w(s)$$

- β_0 is the intercept;
- $x_k(s)$ are covariates included with linear effects;
- β_k are the corresponding coefficients;
- $f_j(\cdot)$ are any non-linear or smooth effects;
- $z_j(s)$ are covariates modelled non-linearly;
- $w(s)$ is the Gaussian random field capturing residual spatial dependence.



SPDE-INLA implementation

4 Methods

Continuous Latent Field

The underlying latent spatial process is continuous and its estimation has a computational cost of n^3 , making classical MCMC approaches computationally demanding.



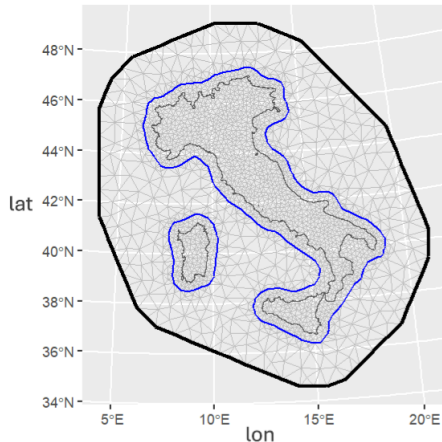
SPDE-INLA approach

The continuous Gaussian Random Field is approximated as a sparse Gaussian Markov Random Field (GMRF) defined on a triangular mesh. INLA provides a computationally efficient alternative based on nested Laplace approximations, providing accurate approximations of posterior distributions.



National-Scale Modelling

Enables feasible and fast spatial modelling at a large national scale.



Chosen triangular mesh over Italy



Selected model

5 Results

Final model structure

$$\begin{aligned} \log \lambda(s) = & \underbrace{\beta_0 + \beta_1 \text{Temp}(s)}_{\text{Fixed effects}} \\ & + \underbrace{f_{\text{elev}}(\text{Elev}(s)) + f_{\text{coast}}(\text{Coast}(s)) + f_{\text{tree}}(\text{Tree}(s)) + f_{\text{popul}}(\text{Popul}(s))}_{\text{Non-linear effects (RW2)}} \\ & + \underbrace{w(s)}_{\text{Latent Gaussian spatial field}}, \quad s \in D \end{aligned}$$

- Temperature: linear effect;
- Elevation, Tree cover density, Distance from coast and Population: RW2 effects;
- $w(s)$: latent Gaussian spatial field.



Fixed effects

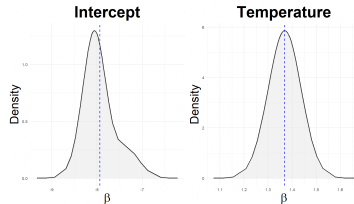
5 Results

Fixed effects describe the global association between covariates and wildfire occurrence intensity, assuming a constant effect across the study area.

- **Temperature** shows a strong positive association with wildfire occurrence intensity;
- The credible interval is above zero, indicating strong posterior support for a positive effect;
- the effect represents a **long-term climatic variable**.

Posterior Statistics for Fixed Effects

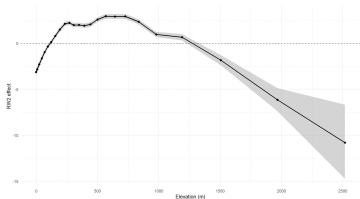
Parameter	Mean	SD	2.5%	50%	97.5%	Mode	KLD
Intercept	-7.940	0.387	-8.573	-7.998	-7.032	-8.038	0
Temperature	1.369	0.068	1.234	1.369	1.502	1.369	0



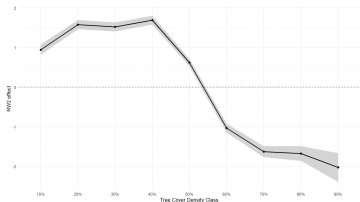


Non-linear Effects: Elevation and Tree cover density

5 Results



Estimated RW2 effect of elevation



Estimated RW2 effect of tree cover density

Elevation

- Highest estimated intensity at intermediate elevations ($\approx 600\text{--}750\text{ m}$);
- lower estimated intensity in lowland and high mountain areas;
- wider credible intervals at the highest elevations.

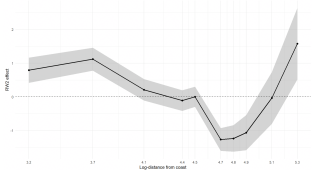
Tree cover density

- Maximum wildfire occurrence intensity at intermediate tree cover ($\approx 20\text{--}40\%$);
- lower intensity in areas with both very sparse and very dense tree cover.

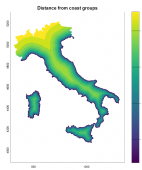


Non-linear Effects: Distance from coast and Population

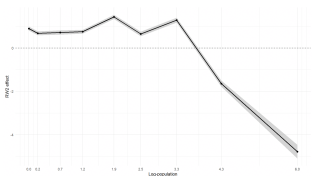
5 Results



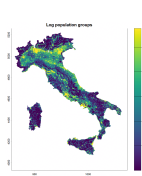
(a) Estimated RW2 effect



(b) Reference raster



(a) Estimated RW2 effect



(b) Reference raster

Distance from the coast

- Higher log-intensity in coastal areas and inlands;
- it acts as a **spatial proxy** summarizing several underlying processes.

Population

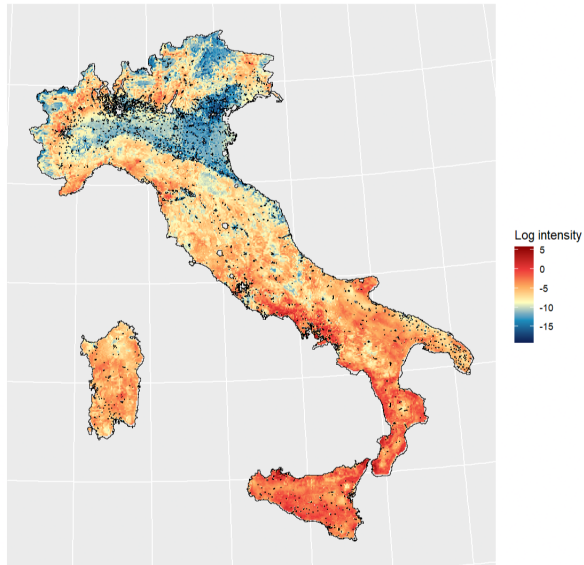
- Higher predicted intensity at low-to-intermediate levels of human presence;
- lower where human presence becomes very high;
- double interpretation.



Estimated Spatial Log-Intensity

5 Results

- Highest estimated intensity in southern and insular Italy;
- lower intensity across most of northern Italy;
- higher log intensity in foothill and pre-Alpine areas;
- local patterns reflect both observed covariates and spatial structure.

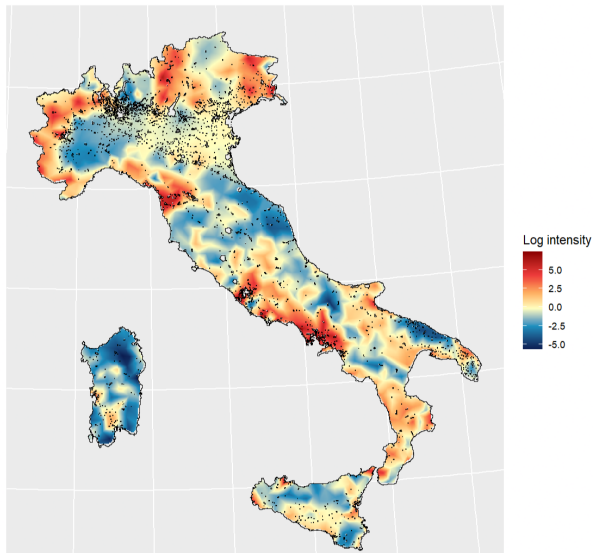




Estimated Spatial Log-Intensity latent gaussian field

5 Results

- The latent field captures residual spatial dependence;
- positive values $\rightarrow$ higher occurrence than expected from covariates;
- negative values $\rightarrow$ lower occurrence than expected;
- this highlights **local-scale unexplained heterogeneity**.





Conclusions and future perspectives

6 Conclusions

The proposed framework provides a reproducible approach to understand spatial patterns of wildfire occurrence and the role of environmental and anthropogenic drivers.

Main contributions

- **Flexible Bayesian spatial framework:**
LGCP–SPDE–INLA to model occurrence intensity with spatial dependence;
- **Reproducible data-driven workflow:**
Harmonized satellite-based products enable consistent national-scale analysis;
- **Insights into drivers:**
Provides concrete insights about environmental and anthropogenic drivers.

Future perspectives

- Spatio-temporal modeling and improved macroclimatic predictors;
- Integration of human accessibility and ignition point dynamics;
- Extension toward fire severity mapping and validation.



Spatial Modelling of Satellite Data for the Development of a Wildfire Intensity Index in Italy

Thank you for your attention.



BAD-ITE

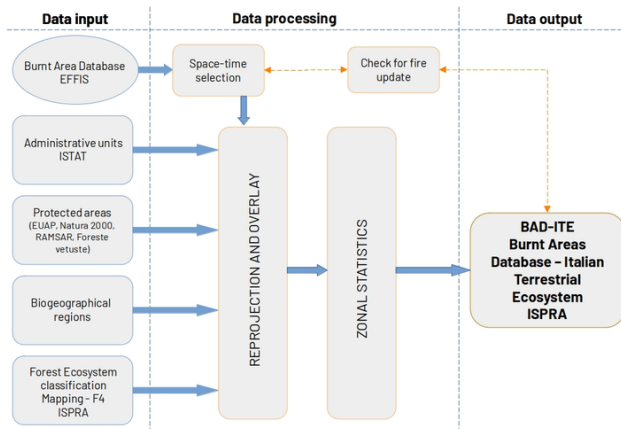
7 Appendix

- The original dataset is derived from EFFIS and it consists of vector polygons delineating burned areas detected from multispectral optical satellite imagery acquired by the MODIS and Sentinel-2 MSI sensors.
- For this analysis it was considered the period that goes from 2018 (year of introduction of Sentinel-2) to 2025 in order to have the same resolution for all the years.
- The original dataset is designed for pan-European monitoring, therefore it lacks information at the national scale. For this reason ISPRA implemented a workflow which consists in overlaying with additional layers like administrative boundaries provided by ISTAT, protected areas layers and forest classifier.



BAD-ITE

7 Appendix





Why LGCP?

7 Appendix

Point Patterns

Goal: Estimate spatial intensity $\lambda(s)$ (expected number of wildfires at location s)



Homogeneous Spatial Poisson Process

Problem: Constant $\lambda(s) = \lambda$ and assumes Complete Spatial Randomness (CSR)



Inhomogeneous Spatial Poisson Process

$\lambda(s)$ not constant, accounts for spatial variation through covariates but **assumes no residual spatial dependence**



Cox Processes

Accounts for unobserved spatial clustering by treating intensity $\lambda(s)$ as a **random variable**



Log-Gaussian Cox Processes (LGCP)

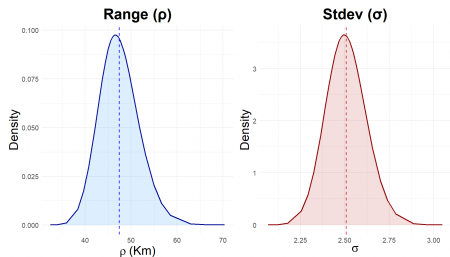
Special case of Cox process where log-intensity $\log \lambda(s)$ is driven by a **Gaussian Random Field**

$$\log \lambda(s) = X^T(s)\beta + \omega(s)$$



Random effects

7 Appendix



Posterior density distributions of SPDE hyperparameters

Random effects are introduced to capture sources of spatial variability that are not explained by the fixed effects.

The latent Gaussian field is characterized by two key hyperparameters:

- **Range (ρ):**
Spatial scale of dependence.
 - Posterior mean = **47.4 km**
- **Marginal SD (σ):**
Magnitude of residual spatial variability.
 - Posterior mean = **2.51**



Model selection criteria

7 Appendix

Goal: identify the LGCP formulation providing the best balance between:

- goodness of fit;
- predictive performance;
- model complexity.

Model selection strategy

1. Screening of correlated predictors;
2. comparison of alternative LGCP formulations;
3. evaluation of linear vs non-linear effects.

Model comparison

- WAIC (primary criterion);
- DIC (additional diagnostic);
- posterior uncertainty of effects;
- stability of covariates effect.



Watanabe–Akaike Information Criterion (WAIC)

7 Appendix

WAIC is a fully Bayesian measure of predictive performance that estimates the **expected out-of-sample predictive accuracy** while accounting for **model complexity**. Unlike DIC, WAIC averages over the full posterior distribution of the model parameters rather than relying on a point estimate.

$$WAIC = -2(lppd - p_{WAIC})$$

where:

- $lppd$ = log pointwise predictive density $\rightarrow$ how well the model predicts the observed data;
- p_{WAIC} = effective number of parameters

Models with lower WAIC values are preferred, as they provide a better trade-off between goodness of fit and predictive performance. In this study, WAIC was adopted as the primary criterion for comparing candidate models.



Deviance Information Criterion (DIC)

7 Appendix

Evaluate the trade-off between model fit and model complexity.

Model deviance: $D(\theta) = -2 \log p(y|\theta)$

Lower deviance indicates a better fit.

Deviance Information Criterion: $DIC = \bar{D} + p_D$

where

$$\bar{D} = E[D(\theta) | y], \quad p_D = \bar{D} - D(\bar{\theta})$$

- $\bar{D}$: posterior mean deviance;
- p_D : effective number of parameters.

Lower DIC $\rightarrow$ better balance between fit and complexity.



Why RW2?

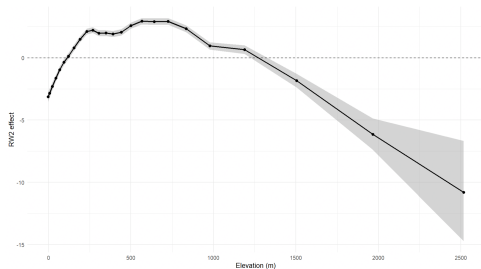
7 Appendix

- non-parametric approach;
- provides a **smoothness constraint** by penalizing rapid changes in the curvature of the function;
- compromise between flexibility and smoothness;
- ideal to model complex or unknown phenomena;
- integrated into the **INLA framework** as a latent Gaussian component.



Non-linear Effects: Elevation

7 Appendix



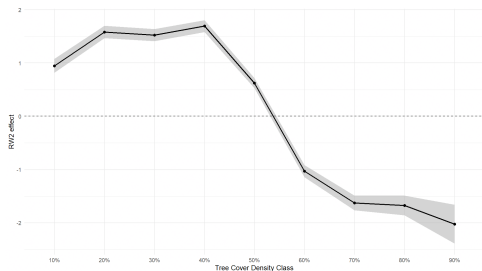
Estimated RW2 effect of Elevation

- Strongly non-linear relationship;
- wildfire occurrence intensity is highest at intermediate elevations ($\approx 600\text{--}750$ m);
- lower estimated intensity in lowland and high mountain areas;
- wider credible intervals at the highest elevations reflect fewer observations.



Non-linear Effects: Tree cover density

7 Appendix



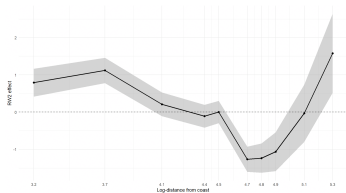
Estimated RW2 effect of Tree cover density

- Strong non-linear relationship;
- maximum wildfire occurrence intensity at intermediate tree cover ($\approx 20-40\%$);
- lower intensity in areas with both very sparse and very dense tree cover;
- may suggest that transitional vegetation landscapes are more prone to wildfire occurrence.

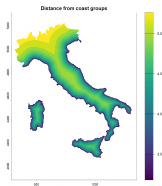


Non-linear Effects: Distance from the Coast

7 Appendix



(a) Estimated RW2 effect



(b) Discretized raster

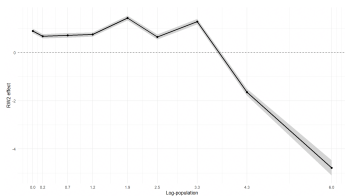
Distance from the coast: posterior non-linear effect (left) and corresponding quantile-based classes (right)

- Highly non-linear relationship;
- higher log-intensity in coastal areas;
- lower intensity at intermediate inland distances;
- slight increase again in the most inland regions, with greater uncertainty;
- it acts as a **spatial proxy** summarizing several underlying processes.

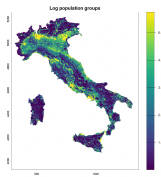


Non-linear Effects: Population

7 Appendix



(a) Estimated RW2 effect



(b) Discretized raster

Population: posterior non-linear effect (left) and corresponding quantile-based classes (right)

- Spatially smoothed to represent surrounding human presence;
- strongly non-linear relationship;
- higher wildfire occurrence at low-to-intermediate levels of human presence;
- lower occurrence where human presence becomes very high;
- double interpretation (higher probability of ignition, but higher landscape fragmentation and more rapid suppression).